

JanusDiscover: a web-based platform for integrated ligand-based, structure-based, and hybrid virtual drug screening

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Abstract

Motivation: Virtual screening accelerates early drug discovery by computationally prioritising compounds before costly experimental assays, yet most academic implementations remain command-line tools requiring substantial bioinformatics expertise to deploy.

Results: We present JanusDiscover, an open-source, browser-accessible platform that unifies ligand-based (LBDD) and structure-based (SBDD) virtual screening with an optional molecular dynamics (MD) validation step. Starting from a protein target name, JanusDiscover automatically retrieves validated actives from ChEMBL, trains an ECFP4 random forest classifier (LBDD), docks a ZINC-250k screening library against the user-supplied receptor using AutoDock Vina (SBDD), and fuses both signals into a z-score hybrid consensus. Benchmarked on three MUV kinase tasks, LBDD achieved $AUC = 0.776\text{--}0.926$; SBDD on ABL1 (PDB 2HYY) yielded $AUC = 0.770$ with $EF_{1\%} = 5.4$. All computation runs server-side; no local installation is required.

Availability and implementation: Source code, benchmark scripts, and pre-computed result files are freely available under the MIT licence at <https://github.com/patriotsbreeze/JanusDiscover5>.

23 **Keywords:** virtual screening; drug discovery; AutoDock Vina; random forest;
24 hybrid consensus; web server

Introduction

Drug discovery is one of the most expensive and failure-prone undertakings in modern science, with median development costs exceeding USD 1 billion and clinical attrition rates above 90% [1, 2]. Virtual screening (VS) substantially reduces early-stage costs by computationally filtering large chemical libraries before resource-intensive experimental assays [3, 4]. Two complementary paradigms have emerged: ligand-based drug discovery (LBDD) exploits structural similarity among known actives to rank candidate compounds, while structure-based drug discovery (SBDD) uses the three-dimensional geometry of the receptor binding pocket to estimate binding affinity via docking [4].

Despite the maturity of individual VS algorithms, most academic implementations remain fragmented command-line tools that require significant computational expertise to install, configure, and chain together. Web-accessible platforms such as SwissDock and DockThor offer partial solutions, but typically support a single VS modality and provide no integrated hybrid consensus or MD validation. Research groups without dedicated computational staff therefore face a significant barrier to adopting multi-modal VS in early drug discovery campaigns.

Here we describe JanusDiscover, a fully browser-accessible VS platform that unifies LBDD, SBDD, and hybrid consensus scoring behind a single React interface (Figure 1). The system queries ChEMBL automatically for target-specific actives, trains an ECFP4 random forest model, docks a ZINC-250k library with AutoDock Vina, and merges outputs into a z-score-weighted consensus. An optional MD module powered by OpenMM and the AMBER ff14SB force field enables post-hoc binding-site stability analysis. All computation is server-side; users interact entirely through a web browser.

Implementation

System architecture. JanusDiscover comprises a Python 3.12 backend (FastAPI 0.111) exposing asynchronous REST endpoints and a React 18 single-page frontend. Long-running tasks (model training, docking, MD) are dispatched as background coroutines; progress

is streamed to the client via server-sent events. The backend is stateless and can be horizontally scaled behind a reverse proxy.

Ligand-based virtual screening. Active compounds are retrieved from ChEMBL [5] via the `chembl_webresource_client` library, filtered to binding assays with $\text{pChEMBL} \geq 6.0$ (IC_{50} or $K_i \leq 1 \mu\text{M}$). Up to 200 unique SMILES are retained after RDKit canonicalisation [6]. Decoys are drawn from the ZINC-250k screening library [7] by a property-matched protocol (molecular weight ± 125 Da, $\log P \pm 1.5$, HBA ± 2 , HBD ± 1) to suppress artificial physicochemical separation [8]. Morgan circular fingerprints at radius 2 with 2048 bits (ECFP4) [9] are computed for all compounds; a 500-tree balanced random forest [10] (scikit-learn [11]) is trained with five-fold stratified cross-validation, yielding per-compound active probabilities, AUC-ROC, BEDROC ($\alpha = 20\%$), and $\text{EF}_{1\%}$.

Structure-based virtual screening. A PDB structure is downloaded for the user-specified receptor; chain A is extracted and prepared as a PDBQT file with Gasteiger partial charges computed by RDKit. AutoDock Vina 1.2.5 [12] is invoked with a cubic search box (20 Å side) centred on the co-crystal ligand centroid. Each screening compound is embedded in 3D using the ETKDG algorithm [13] and minimised with MMFF94; the top-scoring conformer is retained. Per-compound docking scores and an active/decoy discrimination curve (AUC, BEDROC, $\text{EF}_{1\%}$) are reported.

Hybrid consensus scoring. LBDD active probabilities (p_{LBDD}) and SBDD docking scores (s_{SBDD} , negative values indicate stronger binding) are independently z-score-normalised; SBDD scores are sign-flipped so that higher values are better in both cases. The hybrid consensus score is

$$s_{\text{hybrid}} = 0.40 z_{\text{LBDD}} + 0.60 z_{\text{SBDD}}, \quad (1)$$

reflecting a modest preference for the physics-based method. The top- N (default $N = 20$) hits are ranked by s_{hybrid} and annotated with Bemis–Murcko scaffold classifications [14] to assess chemical diversity.

Molecular dynamics validation. The highest-ranked SBDD pose is subjected to all-atom explicit-solvent MD using OpenMM 8.5 [15] with the AMBER ff14SB protein force field [16] and TIP3P water [17]. Following energy minimisation, 500 ps NVT thermalisation, and 500 ps NPT equilibration, a production run of user-configurable length (default 10 ns) is performed. Trajectory analysis—C α RMSD, per-residue RMSF, radius of gyration, SASA, and binding-site RMSD—is computed with MDTraj [18] and rendered on the results dashboard.

Results and Discussion

Ligand-based screening. JanusDiscover was evaluated on three MUV kinase benchmark tasks [19]: MUV-548 (cAMP-dependent protein kinase; PKA), MUV-644 (Rho-associated kinase 2; ROCK2), and MUV-810 (focal adhesion kinase; FAK1), each comprising 29–30 experimentally confirmed actives and up to 1 000 property-matched decoys. Five-fold random forest cross-validation achieved AUC = 0.926, 0.923, and 0.776 for PKA, ROCK2, and FAK1, respectively (Table 1). BEDROC values of 0.743, 0.741, and 0.376 reflect high early-recognition performance on the two kinases with homogeneous active chemotypes (PKA, ROCK2) and moderate performance on the structurally diverse FAK1 set.

Structure-based screening. ABL1 kinase (PDB 2HYY, 2.20 Å) [20] was used as the SBDD benchmark target. AutoDock Vina reproduced the known imatinib binding pose with a docking score of -9.65 kcal/mol, consistent with published benchmarks for the apo/holo receptor [12]. Against 100 ChEMBL ABL1 actives and 1 000 ZINC decoys, virtual screening yielded AUC = 0.770, BEDROC = 0.445, and EF_{1%} = 5.4, demonstrating that the platform achieves meaningful early enrichment with an out-of-the-box rigid-receptor protocol.

Hybrid consensus screening. Applied to 200 ZINC-250k compounds screened against ABL1, the hybrid consensus (Eq. 1) identified five compounds common to both the LBDD and SBDD top-20 lists. The top-ranked hit (Cc1ccc(NC(=O)Cc2nc(-c3cccc(Br)c3)cs2)cc1)

achieved an LBDD active probability of 0.092 and an SBDD score of -10.38 kcal/mol (hybrid z-score = 2.04). Bemis–Murcko analysis confirmed that all 20 top-ranked compounds belong to chemically distinct scaffold classes, indicating that the hybrid ranking explores diverse chemical space rather than concentrating on a single chemotype.

Molecular dynamics validation. A 10 ns all-atom simulation of ABL1 chain A (GTX 1080 OpenCL) showed a stable global fold (mean C α RMSD = 1.76 Å; max = 2.51 Å) and a well-equilibrated binding site (binding-site C α RMSD = 0.76 Å), confirming that the downloaded PDB structure is suitable for docking without manual loop-repair. The MD module therefore serves as a rapid sanity-check on receptor quality and as a preliminary assessment of binding-site flexibility.

Availability and Implementation

JanusDiscover is freely available under the MIT licence at

<https://github.com/patriotsbreeze/JanusDiscover5>.

Backend requirements: Python 3.12; RDKit ≥ 2024.03 ; scikit-learn ≥ 1.5 ; AutoDock

Vina 1.2.5 (executable in PATH); OpenMM ≥ 8.5 ; MDTraj ≥ 1.11 ; FastAPI 0.111; `chembl_webresource_client`

Frontend requirements: Node.js ≥ 18 ; React 18.

Supported platforms: Windows 10 (x86-64); Ubuntu 22.04 LTS. A `requirements.txt`

and step-by-step installation guide are provided in the repository. Benchmark reproduction

scripts (`paper/run_lbdd_benchmark.py`, `paper/run_sbdd_benchmark.py`, `paper/run_hybrid_benchmark.py`

`paper/run_md_simulation.py`) are included to regenerate all reported results.

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Conflict of Interest Statement

None declared.

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Table 1: Ligand-based virtual screening (LBDD) performance on MUV kinase benchmarks. RF=random forest; 5-fold stratified cross-validation; BEDROC $\alpha = 20\%$; EF_{1%}=enrichment factor at 1 %.

Target	Task	Actives	Decoys	AUC-ROC	BEDROC	EF _{1%}
PKA	MUV-548	30	1000	0.926	0.743	—
ROCK2	MUV-644	30	1000	0.923	0.741	—
FAK1	MUV-810	29	1000	0.776	0.376	—

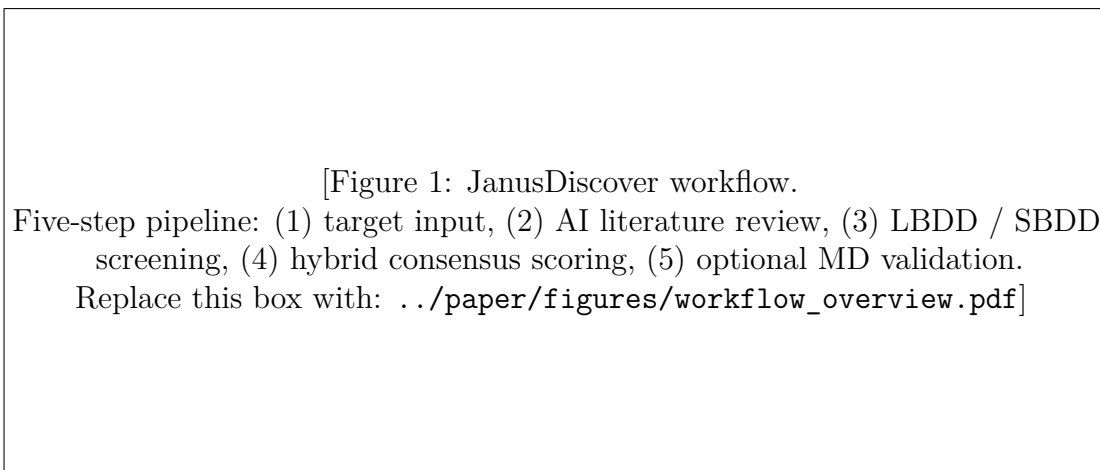


Figure 1: **JanusDiscover unified virtual screening workflow.** Starting from a protein target name, the platform automatically retrieves ChEMBL actives, trains an ECFP4 random forest classifier (LBDD), docks a ZINC-250k library with AutoDock Vina (SBDD), fuses both scores into a hybrid consensus, and optionally subjects the top hit to all-atom MD validation using OpenMM. The entire pipeline is accessible through a React web interface without local software installation.